

Imperial College London

BSc/MSci EXAMINATION May 2023

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Tuesday, 30th May 2023: 14:00 to 16:00

*The paper consists of **three** questions – please answer **all three**.*

A table of Laplace transforms is included at the end of the paper.

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 3 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 3 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. (i) When applied to a sensor, what is meant by each of the terms below?

- (a) Offset
- (b) Saturation
- (c) Loading (include an example)

[7 marks]

(ii) A 'real-world' operational amplifier has a DC gain of 10^5 , an open-loop bandwidth of 10 Hz, and a gain cross-over frequency of 1 MHz. The output phase lag is -90° from 100 Hz to 10^5 Hz and decreases by 45° per decade thereafter.

- (a) Sketch the Bode plot for the amplifier, identifying the phase and gain margins, and hence state whether the amplifier is stable if used with feedback.
- (b) The op-amp is now connected in closed-loop with a feedback fraction $\beta = 0.1$. What is the bandwidth in closed-loop?
- (c) If the op-amp has a DC common-mode gain of 10, what is the DC common-mode rejection ratio (CMRR)? Express your answer in dB.

[9 marks]

(iii) (a) Describe the frequency characteristics of flicker noise.

- (b) Why does the RMS flicker noise in any 'real-world' experiment remain finite even for very low frequency or 'DC' measurements?
- (c) Two identical DC-coupled amplifiers with 10 kHz cut-off frequency have been making a measurement for 24 hours and 5 minutes, respectively. What is the ratio of flicker noise power at their outputs?
- (d) An AC-coupled amplifier has an input resistance of 10 k Ω , a low-frequency limit of 100 Hz, high-frequency limit of 100 kHz, and a noise factor $F = 2$. Assuming thermal noise dominates, what input voltage will give a signal-to-noise ratio (SNR) of 20 dB at the output, at a temperature of 300 K?
- (e) To what temperature must we cool the device to achieve a SNR of 26 dB? [You may need $k_B = 1.38 \times 10^{-23}$ J/K.]
- (f) If 10 V is applied to a forward-biased diode in series with a 10 Ω resistor, what shot noise voltage is measured across the resistor with an instrument of bandwidth 1 MHz? Assume a forward-bias diode voltage of 0.6 V.

[12 marks]

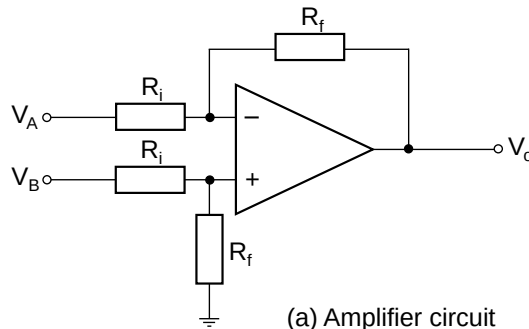
(iv) A 16-bit unipolar digitiser is used to measure a positive voltage signal with a digital resolution of 152.6 μ V and at a rate of 4×10^6 samples per second.

- (a) What is the maximum input signal (in volts) which may be digitised?
- (b) What is the maximum quantisation error?
- (c) What is the RMS voltage noise added to the signal due to quantisation?
- (d) What is the Nyquist frequency for the measurement?
- (e) Which of the frequencies listed here will be properly sampled?
1 MHz; 1.5 MHz; 2.5 MHz; 3 MHz.
- (f) For those improperly sampled, give the alias frequency.

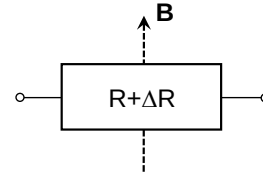
[12 marks]

[Total 40 marks]

2. (i) Diagram (a) below shows an amplifier circuit which uses an operational amplifier which is assumed to have ideal properties. Find an expression for the output of the circuit. [8 marks]



(a) Amplifier circuit



(b) Magnetoresistor

- (ii) Diagram (b) shows a 'magnetoresistor' with resistance $R + \Delta R$ where $R = 1 \text{ k}\Omega$ is the zero-field resistance and ΔR is a positive or negative change in resistance in response to an applied positive or negative magnetic field $\mathbf{B}$ applied in the direction of the arrow. The static sensitivity of the device is defined by $\Delta R = 2 \times 10^5 \text{ }\Omega/\text{T}$, subject to a maximum change in resistance of $\pm 20 \text{ }\Omega$.

- (a) Four magnetoresistors may be arranged as a Wheatstone bridge. Assume a bridge voltage of 5 V. Sketch the circuit and show that the bridge output voltage as a function of applied magnetic field is given by $\Delta V = 1000 B \text{ [V/T]}$. [Hint: assume a particular direction for the field and then consider how you would orient the 4 devices so as to maximise the bridge output.] [8 marks]
- (b) Suggest values for resistors R_f and R_i which will allow circuit (a) to amplify this voltage resulting in an output of 0.65 V at Earth's field (assumed to be $65 \text{ }\mu\text{T}$). [5 marks]
- (c) For the amplified output, sketch the static sensitivity plot as a function of applied magnetic field over the range $\pm 200 \text{ }\mu\text{T}$. Annotate your sketch with any important features. [6 marks]
- (d) The op-amp has, in open loop, a gain $A = 10^5$ and bandwidth of 10 Hz. What will be the bandwidth for a magnetic field measurement? [3 marks]

[Total 30 marks]

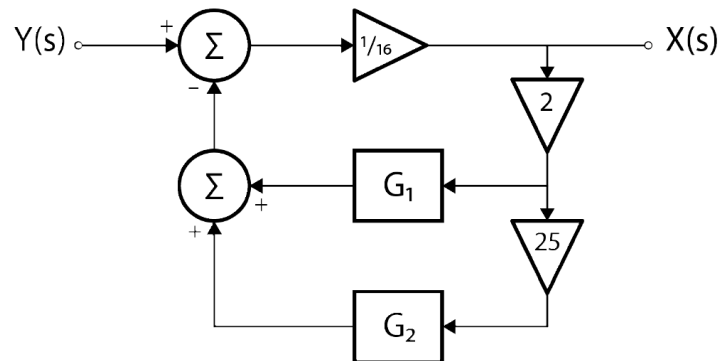
3. (i) Show that the Laplace transform of the unit ramp function $y(t) = t$ is given by:

$$\mathcal{L}\{t\} = \frac{1}{s^2}.$$

[6 marks]

- (ii) Two amplifiers, each having an impulse response $g(t) = e^{-t}$, are connected together in series. A unit ramp function is applied at the input starting at time $t = 0$. Find an expression for the output $x(t)$ for all time $t > 0$. [10 marks]

- (iii) In the diagram below, G_1 is an ideal differentiator and G_2 an ideal integrator.



- (a) Find an expression for the transfer function $G(s)$ of the whole system.

[8 marks]

- (b) For an input $y(t) = 30 u(t)$, show that the output for $t > 0$ is given by:

$$x(t) = 5e^{-4t} \sin 3t.$$

[6 marks]

[Total 30 marks]

You may need this information:

$$\lim_{t \rightarrow \infty} [t e^{-st}] = 0,$$

and there is a table of Laplace transforms overleaf.

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Figure 1: Laplace transforms.